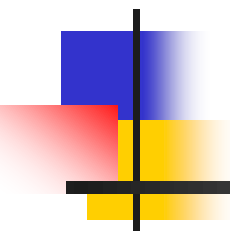


CHAPTER 4:

DISCRETE-TIME FOURIER TRANSFORM

(DTFT)

A decorative graphic is located on the left side of the slide. It features a black crosshair-like structure. The vertical line is intersected by a horizontal line. To the left of the vertical line, there are three overlapping squares: a blue one at the top, a red one in the middle, and a yellow one at the bottom. The horizontal line is a thick, light gray bar that spans across the slide.

Lecture 6: Discrete-Time Fourier Transform (DTFT)

Lecture 7: Digital signal spectra

Lecture 8: Frequency response and filter

Lecture 6

Discrete-Time Fourier Transform (DTFT)

- **Outline:**

1. DTFT basics

2. Inversion of DTFT

3. Properties of DTFT

DTFT formula

$$X(\Omega) = \text{DTFT}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

- Continuous function in frequency and periodic with period of 2π
- Gives the complex frequency spectrum of DT signal
- Not all DTFT is converge

Convergence of the DTFT

We always have:

$$\begin{aligned} \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \right| &\leq \sum_{n=-\infty}^{\infty} \left| x[n] e^{-j\Omega n} \right| \\ &\leq \sum_{n=-\infty}^{\infty} |x[n]| |e^{-j\Omega n}| \\ &= \sum_{n=-\infty}^{\infty} |x[n]| \end{aligned}$$

→ DTFT exists when:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

From ZT to DTFT

Recall ZT of $x[n]$:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Evaluating $X(z)$ on the unit circle (if the unit circle is in the ROC of $X(z)$):

$$X(z) \Big|_{z=e^{j\Omega}} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} = X(\Omega)$$

From ZT to DTFT

DTFT is the Z-transform of $x[n]$ evaluated on the **unit circle**

$$X(\Omega) = X(z) \Big|_{z=e^{j\Omega}}$$

If the ROC of the ZT contains the unit circle, we can get the DTFT from the ZT by substitution **$z = e^{j\Omega}$**

Example of calculating DTFT

Find DTFT of $x(n)$ where $x[n] = a^n u[n]$

If $|a| \geq 1$, DTFT does not exist

If $|a| < 1$:

$$X(\Omega) = \sum_{n=0}^{\infty} a^n e^{-j\Omega n} = \sum_{n=0}^{\infty} (ae^{-j\Omega})^n = \frac{1}{1 - ae^{-j\Omega}} = \frac{e^{j\Omega}}{e^{j\Omega} - a}$$

$$X(z) = \frac{z}{z - a}, \text{ ROC : } |z| > |a| \Rightarrow X(\Omega) = \frac{e^{j\Omega}}{e^{j\Omega} - a}, |a| < 1$$

Lecture 6

DTFT

- **Outline:**

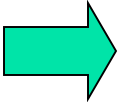
1. DTFT basics

- 2. Inversion of DTFT**

3. Properties of DTFT

Inversion of DTFT

$$\text{————— } X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \text{ —————}$$


$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega l} d\Omega &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \right] e^{j\Omega l} d\Omega \\ &= \sum_{n=-\infty}^{\infty} x[n] \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\Omega(l-n)} d\Omega \right] : \end{aligned}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega n} d\Omega$$

Examples of calculating the inverse DTFT

1. Find $x[n]$ from its DTFT $X(\Omega)$:
$$X(\Omega) = \begin{cases} 1, & |\Omega| \leq \Omega_c \\ 0, & \Omega_c < |\Omega| < \pi \end{cases}$$

$$x[n] = \frac{1}{2\pi} \int_{-\Omega_c}^{\Omega_c} 1 \cdot e^{j\Omega n} d\Omega$$

Examples of calculating the inverse DTFT

2. Find $x(n)$ from its DTFT $X(\Omega)$: $X(\Omega) = \cos^2 \Omega$

Examples of calculating of the inverse of DTFT

3. Find $x(n)$ from its DTFT $X(\Omega)$:
$$X(\Omega) = \frac{e^{j\Omega}}{e^{j\Omega} - 2}$$

Lecture 6

DTFT

- **Outline:**

1. DTFT basics

2. Inversion of DTFT

3. **Properties of DTFT**

Linearity

$$ax[n] + by[n] \xrightarrow{\text{DTFT}} aX(\Omega) + bY(\Omega)$$

The DTFT of a linear combination of two or more signals is equal to the same linear combination of the DTFT of the individual signals.

Time shifting

$$x[n - n_0] \xleftrightarrow{\text{DTFT}} e^{-j\Omega n_0} X(\Omega)$$

Proof: infer from the shifting property of ZT

$$x[n - n_0] \xleftrightarrow{\text{ZT}} z^{-n_0} X(z)$$

→ A shift in time causes **a linear phase shift** in frequency – **no change in DTFT magnitude**

Frequency shifting and modulation

$$e^{j\Omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X(\Omega - \Omega_0)$$

$$\cos(\Omega_0 n) x[n] \xleftrightarrow{\text{DTFT}} \frac{1}{2} X(\Omega - \Omega_0) + \frac{1}{2} X(\Omega + \Omega_0)$$

$$e^{j\Omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} \sum_{n=-\infty}^{\infty} (e^{j\Omega_0 n} x[n]) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j(\Omega - \Omega_0)n} = X(\Omega - \Omega_0)$$

→ Modulation causes **a shift** in frequency

Convolution in time domain

$$x_1[n] * x_2[n] \xleftrightarrow{\text{DTFT}} X_1(\Omega) \cdot X_2(\Omega)$$

Convolution in time $\leftrightarrow$ Multiplication in frequency

Convolution in frequency domain

$$x_1[n] \cdot x_2[n] \xleftrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{2\pi} X_1(\lambda) X_2(\Omega - \lambda) d\lambda = \frac{1}{2\pi} X_1(\Omega) * X_2(\Omega)$$

$$\begin{aligned} x_1[n] \cdot x_2[n] &\xleftrightarrow{\text{DTFT}} \sum_{n=-\infty}^{\infty} (x_1[n] \cdot x_2[n]) e^{-j\Omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{2\pi} X_1(\lambda) e^{j\lambda n} d\lambda \right) x_2[n] e^{-j\Omega n} \\ &= \frac{1}{2\pi} \int_{2\pi} X_1(\lambda) \left(\sum_{n=-\infty}^{\infty} x_2[n] e^{-j(\Omega - \lambda)n} \right) d\lambda \\ &= \frac{1}{2\pi} \int_{2\pi} X_1(\lambda) X_2(\Omega - \lambda) d\lambda \end{aligned}$$

Multiplication in time

Convolution in frequency

HW

Prob.1 Compute the DTFT of the following signals

$$a) \quad x[n] = \left\{ -2, -1, \underset{\uparrow}{0}, 1, 2 \right\}$$

$$b) \quad y[n] = \begin{cases} 2 - (\frac{1}{2})^n & |n| \leq 4 \\ 0 & |n| > 4 \end{cases}$$

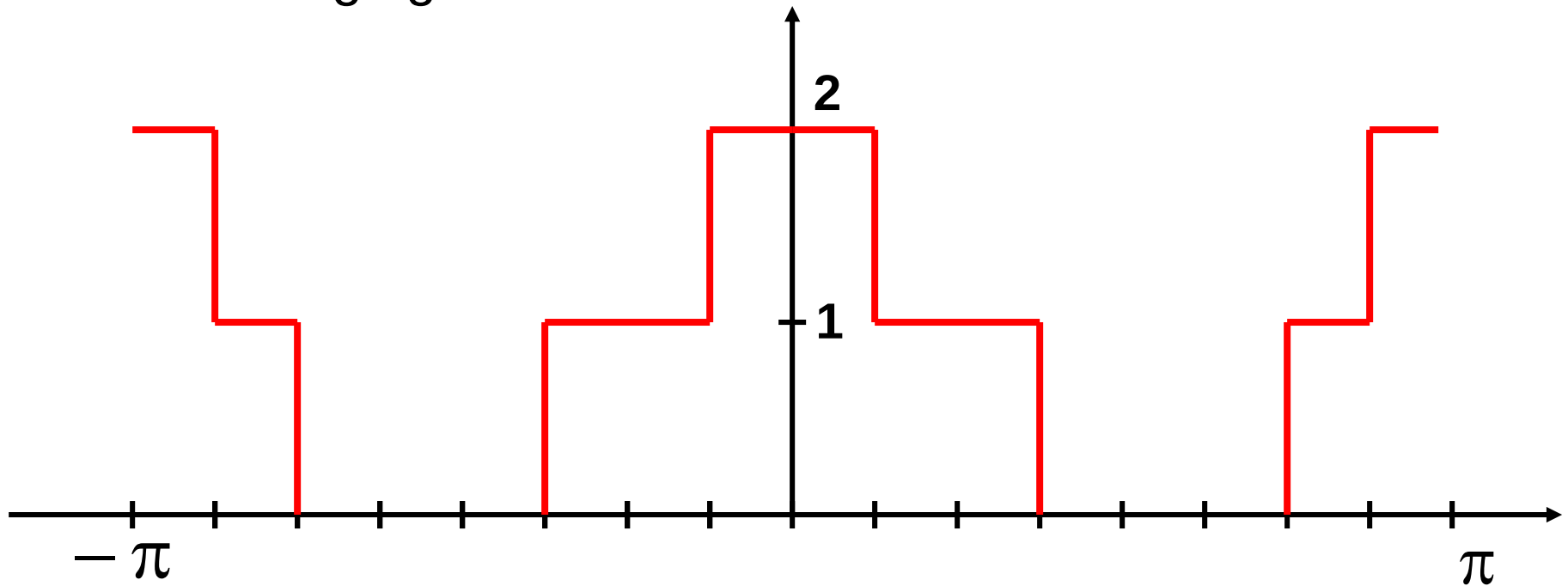
$$c) \quad v[n] = 2^n u[-n]$$

$$d) \quad h[n] = a^n \sin(\Omega_0 n) u[n]$$

$$e) \quad w[n] = u[n] - u[n - 6]$$

HW

Prob.2 Determine the signal having the DTFT shown in following figure



HW

Prob.3 A signal $x(n]$ has the following DTFT:

$$X(\Omega) = \frac{1}{1 - ae^{-j\Omega}}$$

Determine the DTFT of the following signals:

(a) $x[n] * x[n - 1]$

(b) $x[n] \cos(0.3\pi n)$

(c) $e^{j\pi n/2} x[n + 2]$

Lecture 7

Digital signal spectra

- **Outline:**

1. Frequency spectrum

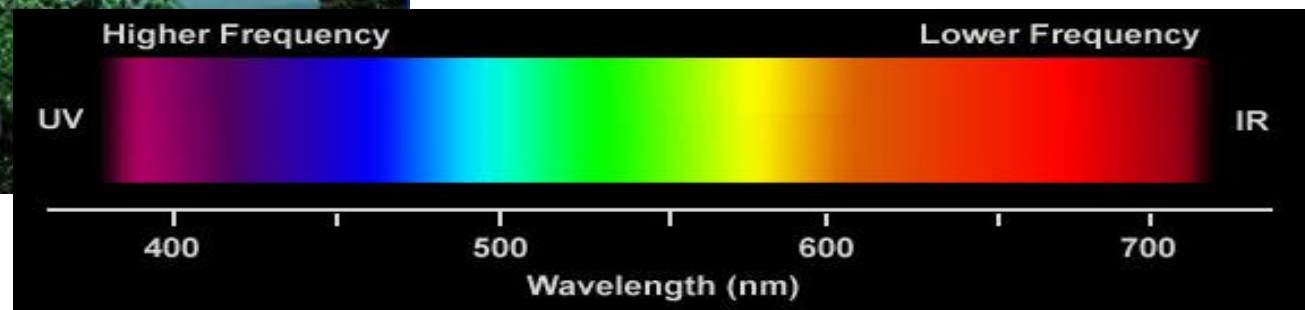
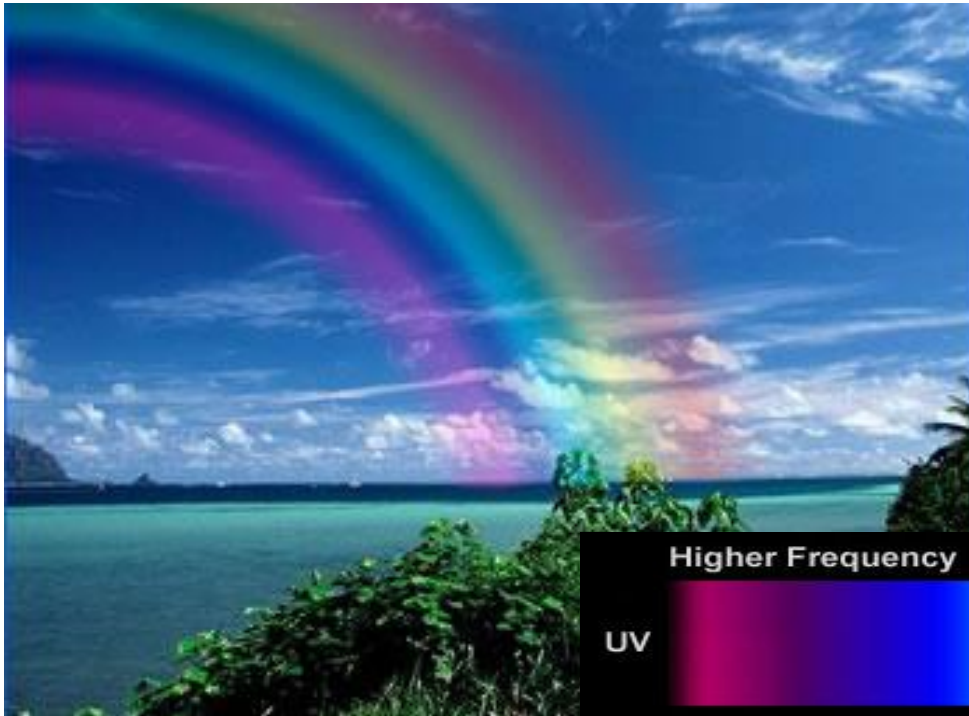
2. Nonperiodic signals

3. Periodic signals

Frequency spectrum

- Detailed description of the frequency components the signal contains.
- **Two parts:** magnitude spectrum & phase spectrum.
- **Tools for calculating the spectrum:** DTFT for nonperiodic signals and DFS for periodic signals.

Frequency spectrum



40-100Hz

1,000Hz - 5,000Hz

10,000Hz - 15,000Hz



**Low Rumble
Of A Truck**



**Sound Of A
Baby Crying**



**The Overtones
Of A High
Pitched Whistle**

Lecture 7

Digital signal spectra

- **Outline:**

1. Frequency spectrum

- 2. Nonperiodic signals**

3. Periodic signals

DTFT to calculate spectrum of a nonperiodic signal

$$X(\Omega) = |X(\Omega)| e^{j\theta(\Omega)}$$

Amplitude spectrum

Phase spectrum

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \quad ; \quad X(-\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{j\Omega n}$$

$$\Rightarrow X(\Omega) = X^*(-\Omega)$$

$$\Rightarrow |X(\Omega)| = |X(-\Omega)| \quad \text{and} \quad \angle X(\Omega) = -\angle X(-\Omega)$$

Example of finding amplitude spectrum and phase spectrum

Find and plot amplitude spectrum and phase spectrum:

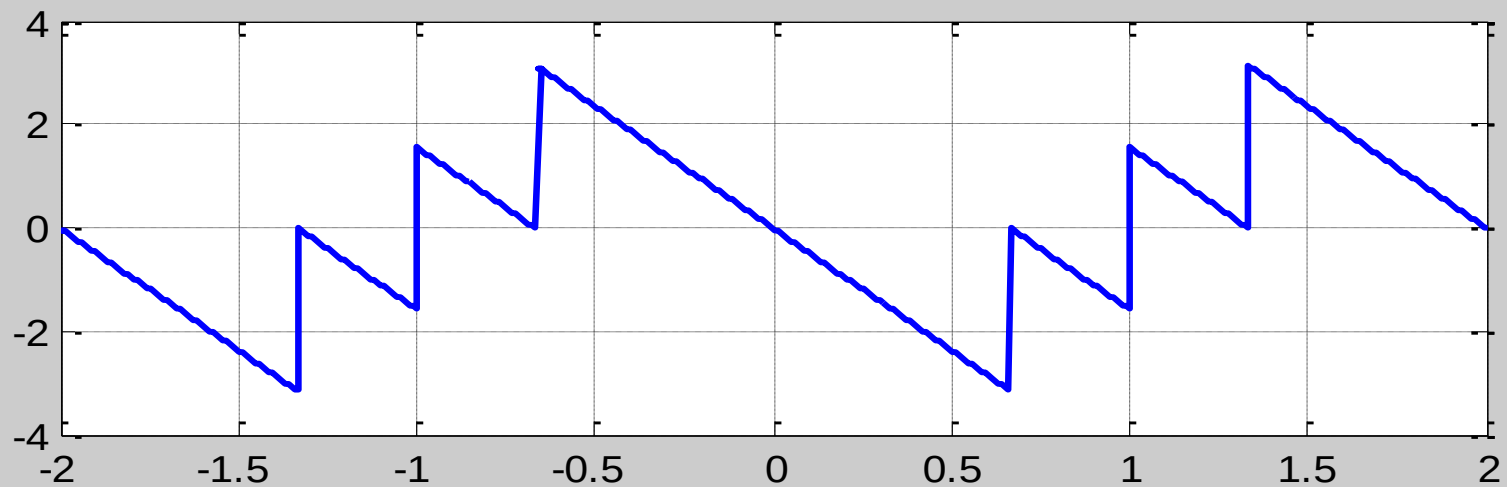
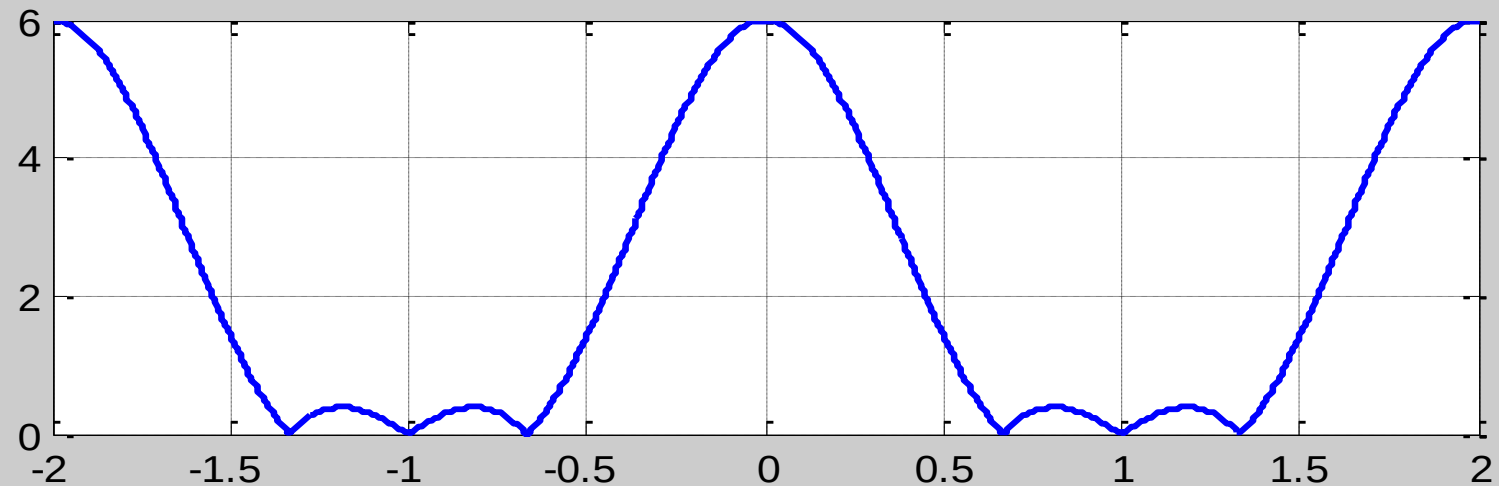
$$x[n] = u[n] - u[n - 4]$$

$$\begin{aligned} X(\Omega) &= \sum_{n=0}^3 e^{-j\Omega n} = \frac{1 - e^{-j4\Omega}}{1 - e^{-j\Omega}} \\ &= \frac{\sin(2\Omega)}{\sin(\Omega/2)} e^{-j3\Omega/2} \end{aligned}$$

Using Matlab to plot amplitude spectrum and phase spectrum

```
w = -2*pi:pi/255:2*pi; % freq.  $-2\pi \rightarrow 2\pi$ , resolution of  $\pi/255$   
X = 4*sinc(2*w/pi)./sinc(w/(2*pi)).*exp(-j*1.5*w);  
subplot(2,1,1);  
plot(w/pi,abs(X)); % plot amplitude spectrum  
subplot(2,1,2);  
plot(w/pi,phase(X)); % plot phase spectrum
```

Amplitude spectrum



Phase spectrum

Lecture 7

Digital signal spectra

- **Outline:**

1. Frequency spectrum

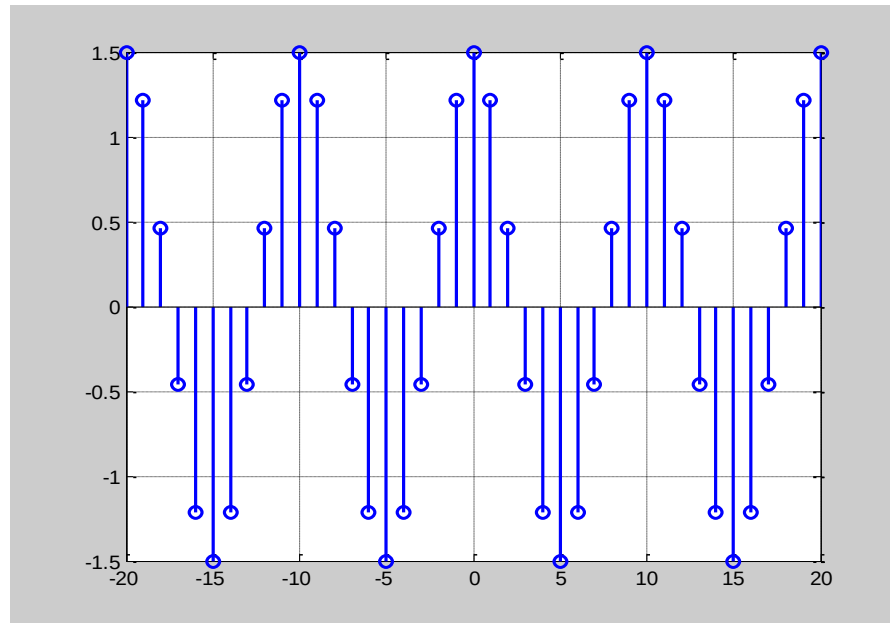
2. Nonperiodic signals

- 3. Periodic signals**

Periodic sinusoidal signals

- Digital sinusoidal signal is periodic only if its frequency F is rational number $x(n) = A \cos(\Omega n + \theta), \quad -\infty < n < +\infty$

$$= A \cos(2\pi F n + \theta), \quad -\infty < n < +\infty$$



Fourier expansion

- CT periodic signals with period T:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}; \quad a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt \quad \omega_0 = \frac{2\pi}{T}$$

- DT periodic signals with period N:

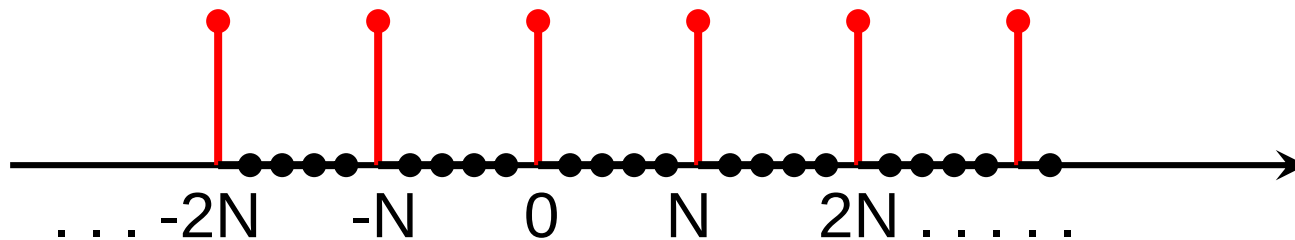
$$x[n] = \sum_{k \in \langle N \rangle} a_k e^{jk\Omega_0 n}; \quad a_k = \frac{1}{N} \sum_{n \in \langle N \rangle} x[n] e^{-jk\Omega_0 n} \quad \Omega_0 = \frac{2\pi}{N}$$

Note: finite sums over an interval length of one periodic **N**

$$e^{jk\Omega_0 n} = e^{jk\frac{2\pi}{N}n} = e^{j(k+N)\frac{2\pi}{N}n} = e^{j(k+N)\Omega_0 n}$$

Example of Fourier Series expansion

Given a DT periodic signals with period N : $p[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$.



$$p[n] = \sum_{k=0}^{N-1} a_k e^{jk \frac{2\pi}{N} n}$$

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} p[n] e^{-jk 2\pi n / N} = \frac{1}{N}$$

Spectrum of periodic signals

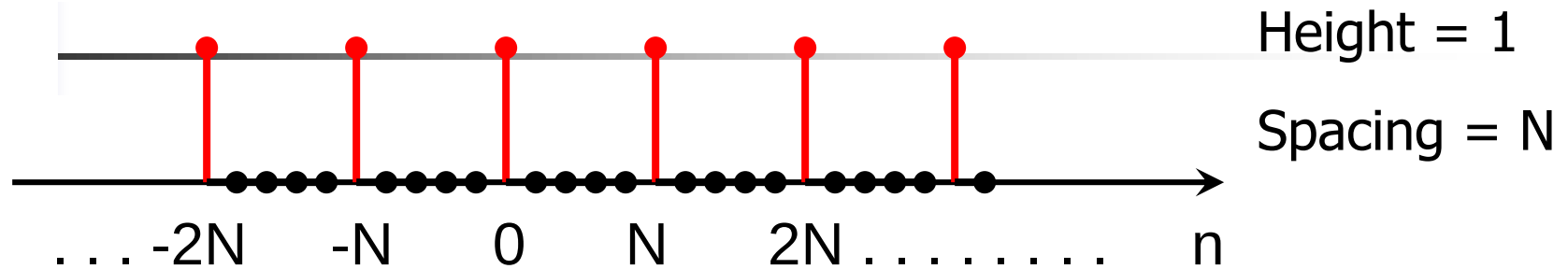
- CT periodic signals with period T:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \xleftrightarrow{F} X(\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$$

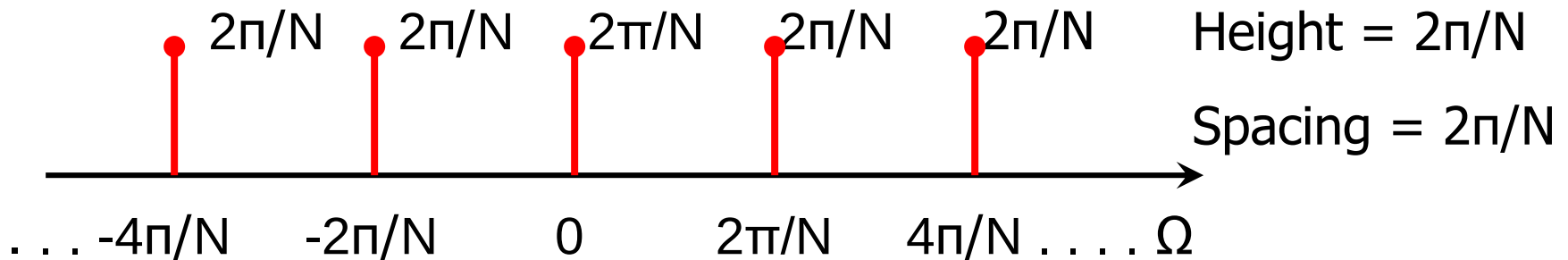
- DT periodic signals with period N:

$$x[n] \xleftrightarrow{F} X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\Omega - k\Omega_0)$$

Example of calculating DTFT of periodic signals (DFS)

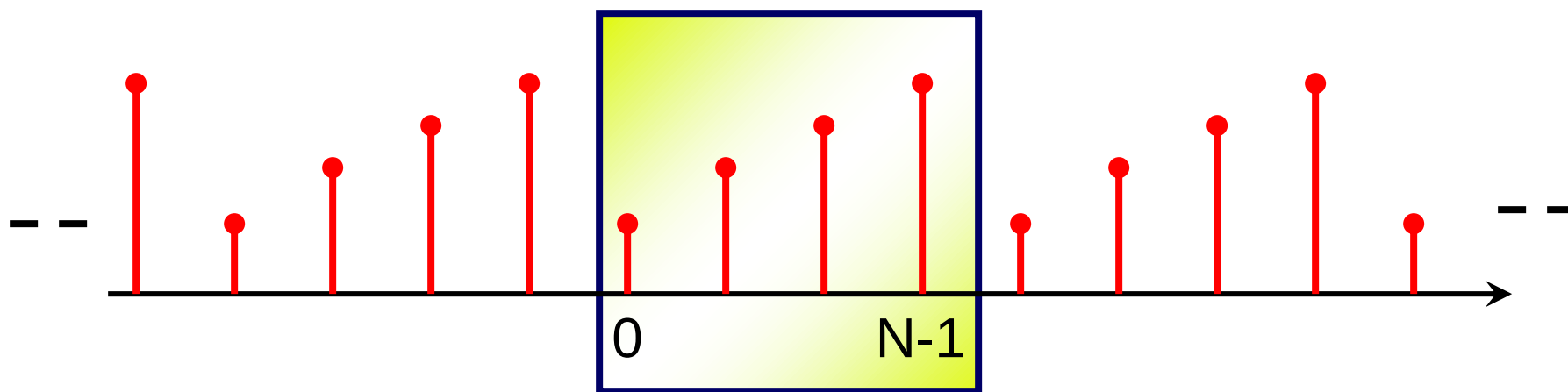


$P(\Omega)$



Another approach to get DTFT of periodic signals (DFS)

$x(n)$ is periodic signal; $x_0(n)$ is a part of $x(n)$ that is repeated



$$x_0[n] = \begin{cases} x[n], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise.} \end{cases}$$

Another approach to get DTFT of periodic signals (DFS)

$$x_0[n] = \begin{cases} x[n], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise.} \end{cases}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x_0[n-kN] = \sum_{k=-\infty}^{\infty} x_0[n] * \delta[n-kN] = x_0[n] * \sum_{k=-\infty}^{\infty} \delta[n-kN]$$

$p(n)$ in previous example

$$x[n] = x_0[n] * p[n] \xrightarrow{F} X_0(\Omega)P(\Omega) = X(\Omega)$$

Another approach to get DTFT of periodic signals (DFS)

$$x[n] = x_0[n] * p[n] \xleftrightarrow{F} X_0(\Omega)P(\Omega) = X(\Omega)$$

$$\begin{aligned} X(\Omega) &= X_0(\Omega) \left(\frac{2\pi}{N} \sum_k \delta\left(\Omega - k \frac{2\pi}{N}\right) \right) \\ &= \frac{2\pi}{N} \sum_k X_0\left(k \frac{2\pi}{N}\right) \delta\left(\Omega - k \frac{2\pi}{N}\right) \end{aligned}$$

N samples—you are sampling the DTFT at N equal intervals around the unit circle

→ It has **N** distinct values at $k = 0, 1, \dots, N-1$

Procedure to calculate spectrum (DFS) of periodic signals

Step 1:

Start with $x_0(n)$ – one period of $x(n)$, with zero everywhere else

Step 2:

Find the DTFT $X_0(\Omega)$ of the signal $x_0[n]$ above

Step 3:

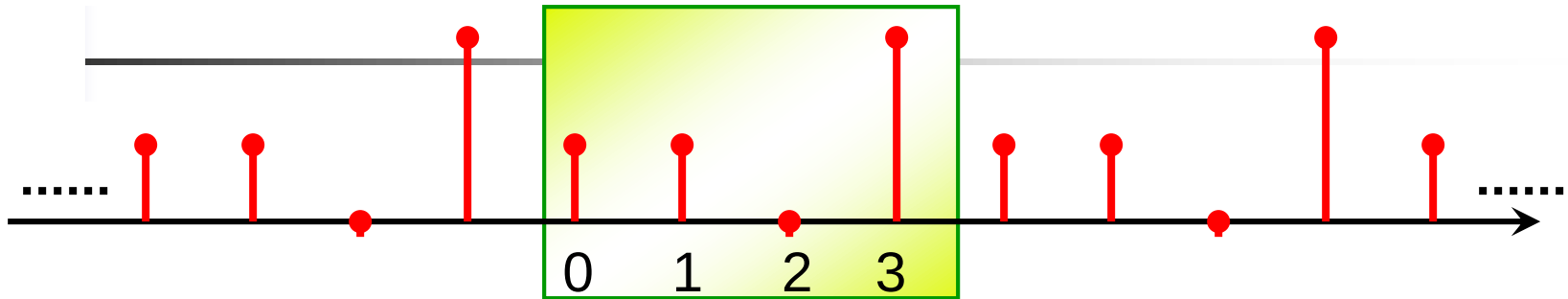
Find $X_0(\Omega)$ at N equally spacing frequency points $X_0(k2\pi/N)$

Step 4:

Obtain the DTFT(DFS):

$$X(\Omega) = \frac{2\pi}{N} \sum_k X_0\left(k \frac{2\pi}{N}\right) \delta\left(\Omega - k \frac{2\pi}{N}\right)$$

Example of calculating DFS of periodic signals



$$X_0(\Omega) = \sum_{n=0}^3 x_0(n)e^{-j\Omega n} = 1 + e^{-j\Omega} + 2e^{-j3\Omega}$$

$$X_o(\frac{2\pi k}{4}) = 1 + e^{-j\frac{2\pi k}{4}} + 2e^{-3j\frac{2\pi k}{4}} \quad k = 0, 1, 2, 3$$

$$k = 0 \rightarrow X_0(0) = 4; \quad k = 1 \rightarrow X_0(1) = 1+j$$

$$k = 2 \rightarrow X_0(2) = -2; \quad k = 3 \rightarrow X_0(3) = 1-j$$

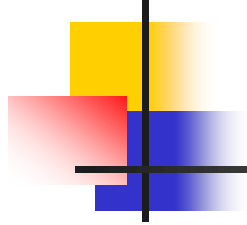
Example of calculating DFS of periodic signals

$$\begin{aligned} X(\Omega) &= \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} X_o\left(\frac{2\pi k}{N}\right) \delta\left(\Omega - \frac{2\pi k}{N}\right) \\ &= \frac{2\pi}{4} \sum_{k=-\infty}^{\infty} X_o\left(\frac{2\pi k}{4}\right) \delta\left(\Omega - \frac{2\pi k}{4}\right) \end{aligned}$$

For one period $0 \leq \Omega < 2\pi$

$$\frac{\pi}{2} \left\{ 4\delta(\Omega) + (1+j)\delta\left(\Omega - \frac{\pi}{2}\right) - 2\delta(\Omega - \pi) + (1-j)\delta\left(\Omega - \frac{3\pi}{2}\right) \right\}$$

HW



Prob.4 Determine and sketch the magnitude and phase response of the following signals

$$a) \quad x[n] = \frac{1}{2}(\delta[n] - \delta[n-2])$$

$$b) \quad x[n] = \frac{1}{2}(\delta[n] - 2\delta[n-1] + \delta[n-2])$$

$$c) \quad x[n] = \frac{1}{8}(\delta[n] + 3\delta[n-1] + 3\delta[n-2] + \delta[n-3])$$

HW

Prob.5

Given a periodic signal $y(n)$ with $N = 3$ with associated

$$y_0[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

Find $Y_0(\Omega)$ and $Y(\Omega)$. Do the test with the inverse DTFT

Lecture 8

Frequency response and filter

- **Outline:**

1. Frequency response

2. Response to complex exponential signals

3. Filters

Frequency response

- For impulse response, $h(n)$, its DTFT is often called **frequency response $H(\Omega)$**
- **$H(\Omega)$** completely characterizes a LTI system in the frequency domain
- **$H(\Omega)$** allows us to determine the steady-state response of the system to any arbitrary weighted linear combination of sinusoids or complex exponential

Example of frequency response

- A LTI causal system is described by the following equation:

$$y[n] + 0.1y[n-1] + 0.85y[n-2] = x[n] - 0.3x[n-1]$$

- First, checking the stability of the system (by using Matlab):

`b = [1 -0.3];`

`a = [1 0.1 0.85];`

`zplane(b,a)` *% plot zeros and poles to check if all poles are inside the unit circle*

- Second, take DTFT for two sides:

$$H(\Omega) = \frac{1 - 0.3e^{-j\Omega}}{1 + 0.1e^{-j\Omega} + 0.85e^{-j2\Omega}}$$

Example of amplitude & phase responses

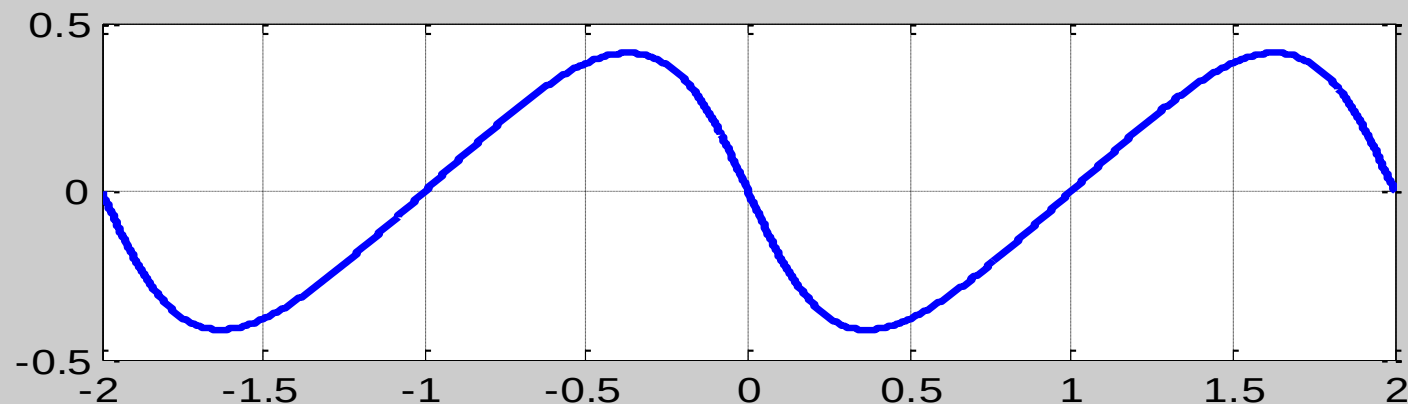
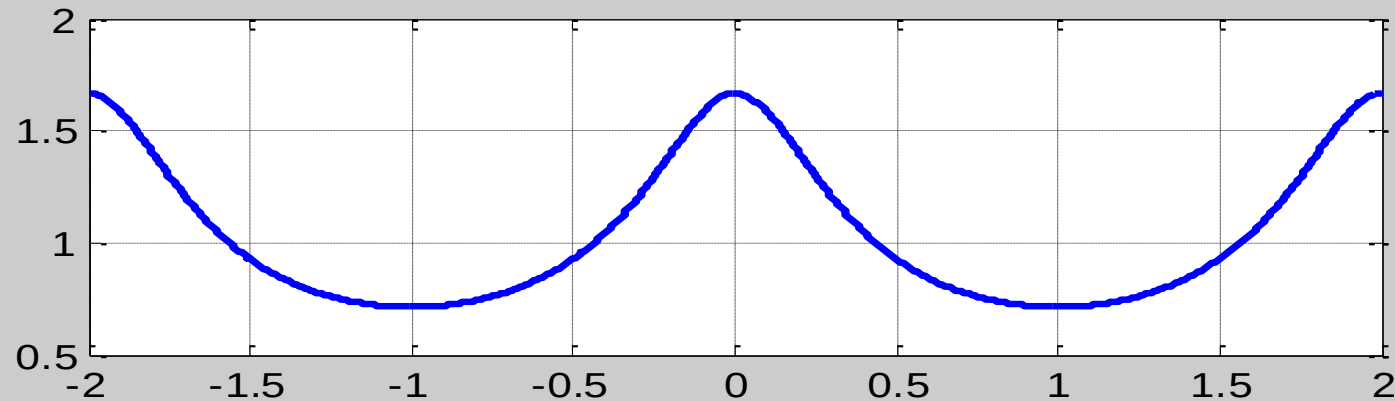
$$\underline{\hspace{10em}} H(\Omega) = \frac{1}{1 - 0.4e^{-j\Omega}} \underline{\hspace{10em}}$$

$$\begin{aligned} |H(\Omega)| &= \frac{1}{|1 - 0.4e^{-j\Omega}|} = \frac{1}{|1 - 0.4\cos\Omega + j0.4\sin\Omega|} \\ &= \frac{1}{\sqrt{(1 - 0.4\cos\Omega)^2 + (0.4\sin\Omega)^2}} \\ &= \frac{1}{\sqrt{1.16 - 0.8\cos\Omega}} \end{aligned}$$

$$\angle H(\Omega) = 0 - \angle(1 - 0.4e^{-j\Omega}) = -\operatorname{arctg}\left(\frac{0.4\sin\Omega}{1 - 0.4\cos\Omega}\right)$$

Example of amplitude & phase responses

$$H(\Omega) = \frac{1}{1 - 0.4e^{-j\Omega}}$$



Lecture 8

Frequency response and filter

- **Outline:**

1. Frequency response

2. Response to complex exponential signals

3. Filters

Response to complex exponential signals

$$x[n] = Ae^{j\Omega_0 n}, \quad -\infty < n < \infty$$


$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]$$

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k] \left(Ae^{j\Omega_0 (n-k)} \right) \\ &= A \left[\sum_{k=-\infty}^{\infty} h[k] \left(e^{-j\Omega_0 k} \right) \right] e^{j\Omega_0 n} \\ &= (Ae^{j\Omega_0 n}) H(\Omega_0) = x[n] H(\Omega_0) \end{aligned}$$

Example of determining response to complex exponential signals

Determine the output signal of system $h[n] = (1/2)^n u[n]$ to the input signal below

$$x[n] = A e^{j\frac{\pi}{2}n}, \quad -\infty < n < \infty$$

$$H\left(\frac{\pi}{2}\right) = \frac{1}{1 + j\frac{1}{2}} = \frac{2}{\sqrt{5}} e^{-j26.6^\circ}$$

$$y[n] =$$

Response to sinusoidal signals

$$x[n] = A \cos(\Omega_0 n) = \frac{A}{2} e^{j\Omega_0 n} + \frac{A}{2} e^{-j\Omega_0 n}, \quad -\infty < n < \infty$$

$$y[n] = \frac{A}{2} e^{j\Omega_0 n} H(\Omega_0) + \frac{A}{2} e^{-j\Omega_0 n} H(-\Omega_0)$$

$$= \frac{A}{2} e^{j\Omega_0 n} |H(\Omega_0)| e^{j\angle H(\Omega_0)} + \frac{A}{2} e^{-j\Omega_0 n} |H(\Omega_0)| e^{-j\angle H(\Omega_0)}$$

$$= A |H(\Omega_0)| \cos(\Omega_0 n + \angle H(\Omega_0))$$

Example of determining response to sinusoidal signals

$$h[n] = (1/2)^n u[n]$$

$$x[n] = 10 - 5 \sin \frac{\pi}{2} n + 20 \cos \pi n, \quad -\infty < n < \infty$$

$$H(0) = \frac{1}{1-0.5} = 2; H\left(\frac{\pi}{2}\right) = \frac{2}{\sqrt{5}} e^{-j26.6^\circ}; H(\pi) = \frac{1}{1+0.5} = \frac{2}{3}$$

$$y[n] = 10 |H(0)| - 5 \left| H\left(\frac{\pi}{2}\right) \right| \sin\left(\frac{\pi}{2} n + \angle H\left(\frac{\pi}{2}\right)\right)$$

$$+ 20 |H(\pi)| \cos(\pi n)$$

$$= 20 - 2\sqrt{5} \sin\left(\frac{\pi}{2} n - 26.6^\circ\right) + \frac{40}{3} \cos(\pi n)$$

Lecture 8

Frequency response and filter

- **Outline:**

1. Frequency response

2. Response to complex exponential signals

- 3. Filters**

What is digital filters?

- Systems that perform mathematical operations on a DT signal to reduce or enhance certain aspects of that signal.
- Provide a convenient means to change the nature of a signal.
- **Change the frequency characteristics** of a signal in a specific way, letting some frequencies in the signal pass while blocking other

B
E
F
O
R
E



A
F
T
E
R

Characterizations of digital filter

1. Impulse response $h(n)$: FIR and IIR filters
2. Transfer function $H(z)$
3. Frequency response $H(\Omega)$: amplitude response and phase response
4. Difference equation:

$$y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

Typical applications of digital filters

- **Noise Suppression:** radio signal, biomedical signal, analog media signal...
- **Enhancement of Selected Frequency Range:** treble/bass control, equalizers in audio, image edge enhancement...
- **Bandwidth Limiting:** aliasing prevention, interference avoidance...
- **Removal of Specific Frequencies:** DC removal, 60 Hz signal removal, notch filter...
- **Special Operations:** differentiation, integration, Hilbert transform...

Basic digital filter types

- **Low pass filter (LPF):** lets low frequencies through while blocking high frequencies;
- **High pass filter (HPF):** lets high frequencies while blocking low frequencies;
- **Band pass filter (BPF):** allows a band of frequencies to pass;
- **Band stop filter (BSF):** allows all frequencies outside a band to pass

Ideal digital filters

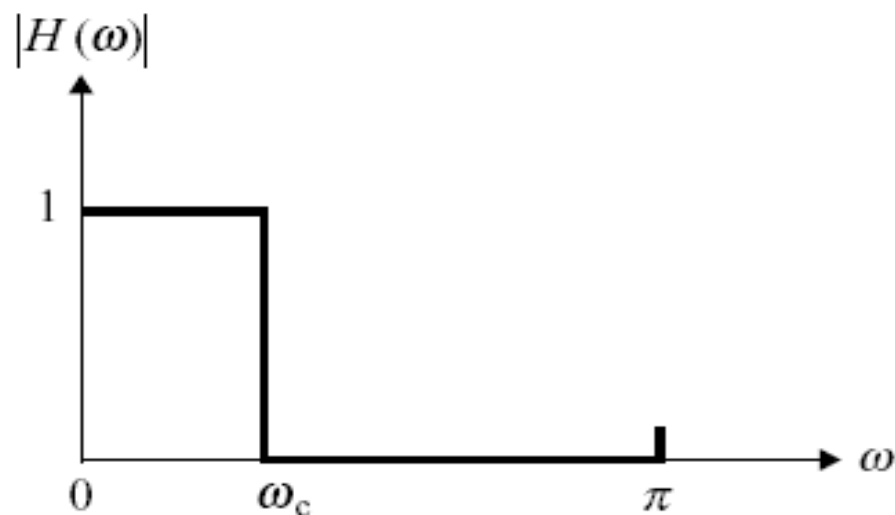
$$y[n] = \begin{cases} Cx[n - n_0], & \omega_1 < \omega < \omega_2 \\ 0, & \omega \neq \end{cases}$$

⇒ $Y(e^{j\omega}) = CX(e^{j\omega})e^{-j\omega n_0} = X(e^{j\omega})H(e^{j\omega}), \quad \omega_1 < \omega < \omega_2$

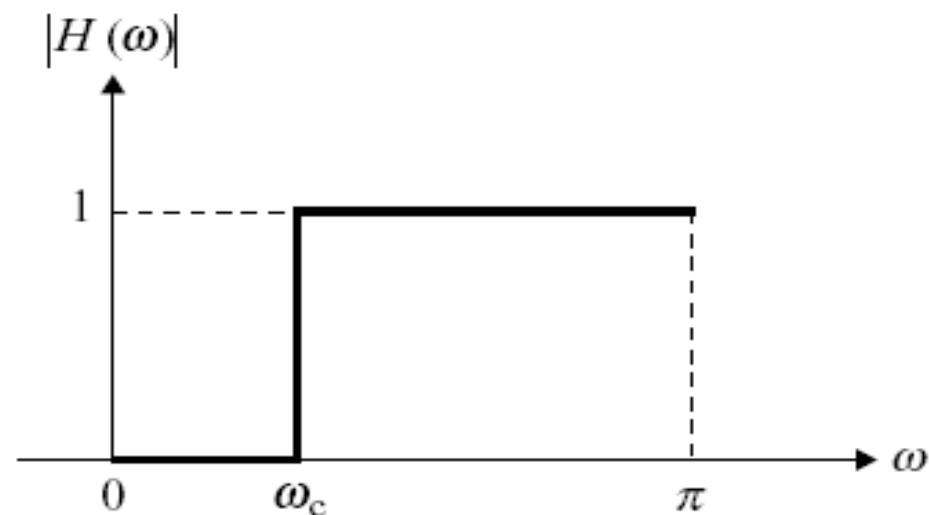
⇒ $H(e^{j\omega}) = \begin{cases} Ce^{-j\omega n_0}, & \omega_1 < \omega < \omega_2 \\ 0, & \omega \neq \end{cases}$

⇒ $|H(e^{j\omega})| = C, \quad \omega_1 < \omega < \omega_2$ Linear phase response
 $\theta(\omega) = -\omega n_0, \quad \omega_1 < \omega < \omega_2$ Constant amplitude response

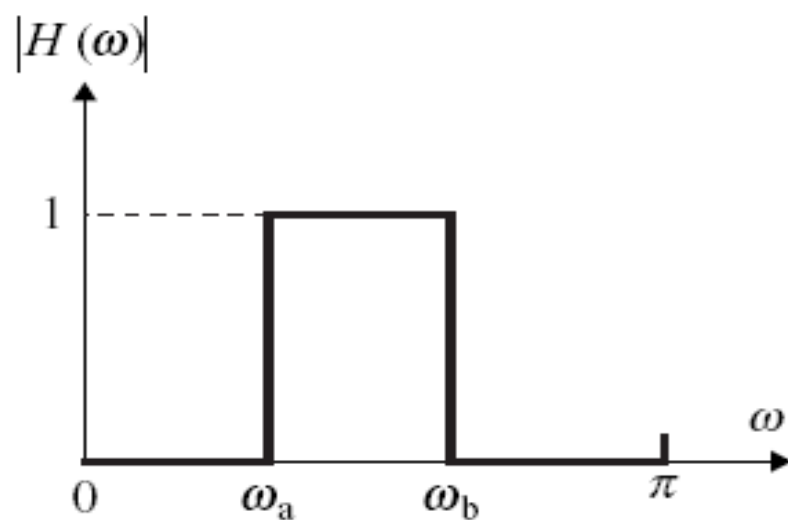
Ideal digital filters



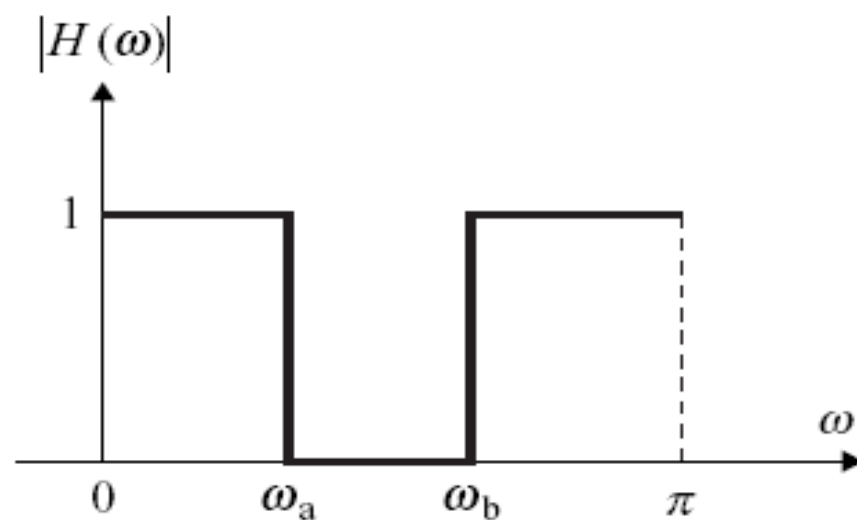
(a) Lowpass filter



(b) Highpass filter



(c) Bandpass filter



(d) Bandstop filter

Block diagram representation

- Being built based on input-output relation or transform function by connecting the basic components in the specific way:

$$y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

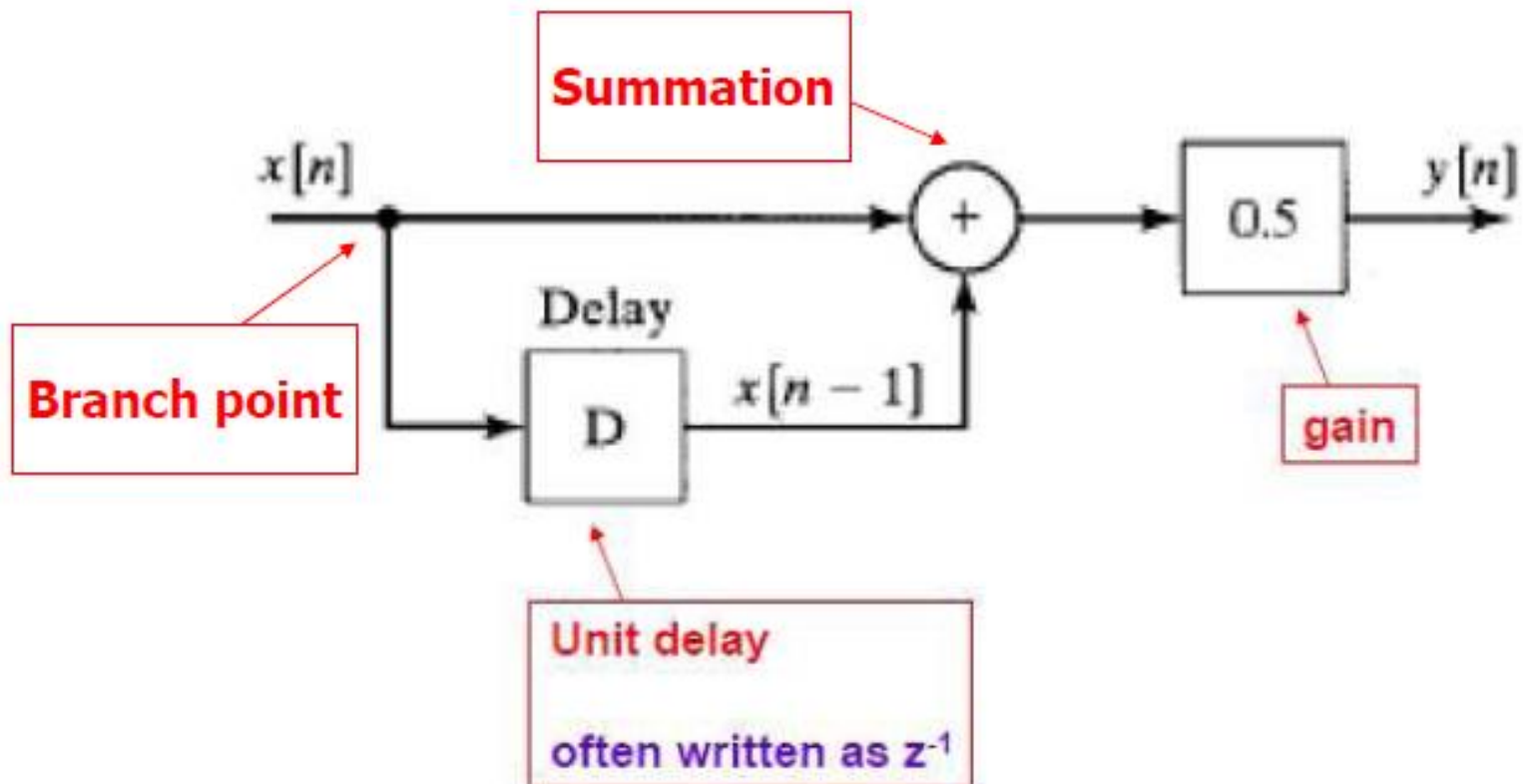
$$\Leftrightarrow \sum_{k=0}^N a_k y[n-k] = \sum_{r=0}^M b_r x[n-r]$$

$$\Leftrightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

- Some advantages:** easy to determine the relation between input and output, easy to manipulate the diagram to the other equivalent one, easy to determine the hardware...

Components for block diagram

EX: an averaging system $y[n] = 0.5(x[n] + x[n-1])$



HW

Prob.6 An FIR filter is described by the difference equation

$$y[n] = x[n] - x[n-10]$$

- a) Compute and sketch its magnitude and phase response
- b) Determine its response to the inputs

$$(1) \quad x[n] = \cos \frac{\pi}{10} n + 3 \sin \left(\frac{\pi}{3} n + \frac{\pi}{10} \right) \quad -\infty < n < \infty$$

$$(2) \quad x[n] = 10 + 5 \cos \left(\frac{2\pi}{5} n + \frac{\pi}{2} \right) \quad -\infty < n < \infty$$

HW

Prob.7 (a) Determine the coefficients of a linear-phase FIR filter

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

such that:

- It rejects completely a frequency component at $\Omega_0 = 2\pi/3$
- Its frequency response is normalized so that $H(0) = 1$

(b) Compute and sketch the magnitude and phase response of the filter to check if it satisfies the requirements

- $w = -\pi:\pi/255:\pi;$
- $b_0 = 1/3$; $b_1 = 1/3$; $b_2 = 1/3$;
- $H = b_0 + b_1 \cdot \exp(-j \cdot w) + b_2 \cdot \exp(-j \cdot 2 \cdot w);$
- $H_{abs} = \text{abs}(H);$
- $H_{phase} = \text{phase}(H);$
- `subplot(2,1,1);`
- `plot(w/pi,Habs,'linewidth',2);grid;`
- `subplot(2,1,2);`
- `plot(w/pi,Hphase,'linewidth',2);grid;`